

Andrea Carvajal Maldonado

andreacarvajal0419@gmail.com



+57 3155777068



[LinkedIn](#)

SUMMARY

Bilingual mathematical engineer with strong motivation for data analysis, statistics and Machine Learning. Demonstrated ability in mathematical logical thinking and problem solving. Main skills include working in interdisciplinary teams, effective communication and organization. Experience in programming languages such as Python, R and Matlab. SQL and Power BI fundamentals. Key strengths in leadership and group and event management.

EDUCATION

EAFIT University, Medellín

Bachelor of Mathematical Engineering, Minor in Computer Science

June 2019 - June 2024

GPA: (4.42 / 0.0-5.0)

EXPERIENCE

Intern at IBM Colombia

Jan 2023 – Dec 2023

- Generate reports on server performance for various clients
- Manage internal databases and aim to optimize efficiency and accessibility
- Analyze server data and present insights to aid in decision-making processes
- Contribute to the automation of human resources processes

Data Analyst at Vensure Employer Solutions

Jan 2025 – Present

- Clean and preprocess raw data to eliminate inconsistencies, errors, and inaccuracies.
- Validate data integrity through rigorous quality checks and validation processes.
- Design and maintain efficient database structures for storing structured and unstructured data.
- Organize data logically and coherently, facilitating easy retrieval and analysis
- Perform data transformations, including aggregation and normalization, to prepare data for analysis
- Generate regular reports, dashboards, and visualizations to communicate data-driven insights.
- Create visual representations that aid in understanding trends, patterns, and key metrics

KEY PROJECTS

Prediction of Colombian Students' Academic Success Using Decision Trees Algorithms

Sep 2020 – Nov 2020

Python

- Implemented the CART algorithm from scratch to predict the performance of Colombian students in the Saber Pro national standardized test based on data from their Saber 11.
- Obtained an average AUC of 0.79 and an average precision score of 0.81

Robust Logistic Regression

Sep 2021 – Nov 2021

Python

- Computed a robust covariance matrix using the Kendall and Spearman correlation coefficients.
- Created a robust version of the logistic regression using the robust covariance matrix.

Mesh Optimization Using NSGA-II

Sep 2021 – Nov 2021

Python

- Implemented a multi-objective genetic algorithm from scratch to optimize 3D meshes.
- Minimized the meshes' approximation error and the standard deviation of their triangles.

Fire Detection Using Computer Vision

Jan 2022 - Jun 2022

Python-Pytorch

- Implemented the instance segmentation algorithm to classify fire and non-fire images.

Functional data classification methods

Jan 2022 - Present

R-fda

- Implement a statistical supervised technique to classify a set of functional data into two or more groups.

RELEVANT COURSES

Optimization, Heuristics, Data Structures and Algorithms, First Order Logic, Discrete Mathematics, Probability theory, Multivariate Statistics, Advanced Multivariate Statistics for Machine Learning, Robust and non-parametric statistical techniques, Stochastic Processes, Artificial Intelligence.

TECHNICAL SKILLS

Programming Languages Languages	Fluent: Python. Spanish (Native), English (Advanced), French (Basic)	Experience in: Java, MATLAB, R, Pytorch, Sklearn, SQL, Power BI.
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EXTRACURRICULAR ACTIVITIES

Human Resources Director of EN FEMENINO	<i>Jan 2023 – Dec 2023</i>
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- Promoted dialogue regarding gender equity and ensure member well-being.

General Director of mathematical engineering student committee CIGMA-OE	<i>Jan 2023 – Dec 2023</i>
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- Led academic and integration events for 30+ students.

Human Resources Director of mathematical engineering student committee CIGMA-OE	<i>Jan 2022 – Dec 2022</i>
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- Managed training, welfare, and recruitment of members.

Board of directors of DÍAS EAFIT	<i>Jan 2022 – Dec 2022</i>
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- Planned and coordinated one of the university's largest cultural events.

Volunteer in subsidies program at Organización Estudiantil – EAFIT	<i>Jul 2022 – Jun 2024</i>
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- Assessed applicants for food and transport subsidies.